DEEP LEARNING FOR FINANCE

Project # 3

Tasks

Step 1

a. Gather information on the time series of the prices of any security of your choice: equity, cryptocurrencies, options, bonds, volatilities… Be sure to include a graphical and textual description of the data used.

b. You will build a predictive model to forecast the returns (or volatilities or yield changes, etc.) of the security you have chosen. Build the labels of the model so that it is apparent that there exists some leakage of information between any choice of the training and test samples in the predictive model.

c. Use three DL models to predict the time series, using a single train/test split. The three models should be:
    a. 1) an MLP,
    b. 2) an LSTM, and
    c. 3) a CNN based on GAF.

d. Provide information of the results from backtests of the trading strategies that arise from each of the three models.

Note: The time series should not exceed **2,000 observations**. You can set the model as regression or classification problems (or both).

Step 2

Now you will backtest the model using a walk forward method.

a. Train your models using a (non-anchored) walk forward method that exploits a train/test split with 500 observations in each set.

b. Train your models using a (non-anchored) walk forward method that exploits a train/test split with 500 observations in each training set and 100 observations in the test sets.

c. Discuss how the results of the backtests in parts a and b change with respect to those in Step 1.

d. Discuss how the results of the backtests compare between parts a and b. Can backtest overfitting due to leakage explain the results?

Step 3

Now you are going to repeat the same process as in Step 2 but alleviating the leakage of information between training and test sets.

a. Set up and describe a method to reduce the extent of leakage between training and test samples.

b. Using your method to reduce leakage, train your models using a (non-anchored) walk forward method that exploits a train/test split with 500 observations in each set.

c. Using your method to reduce leakage, train your models using a (non-anchored) walk forward method that exploits a train/test split with 500 observations in each training set and 100 observations in the test sets.

d. Discuss how the results of the backtests compare between parts b and c. Has overfitting apparently disappeared, as compared to Step 3?

Step 4

As a team, the group will organize both the Python file for all the steps as well as work together on a report that contains the answers to all the questions above (except for the code) in a clear and organized manner (think about how you would present this to your boss). Be sure that the question number appears in the Python notebook before your response. For example, *"Step 3a. We describe a method …"*